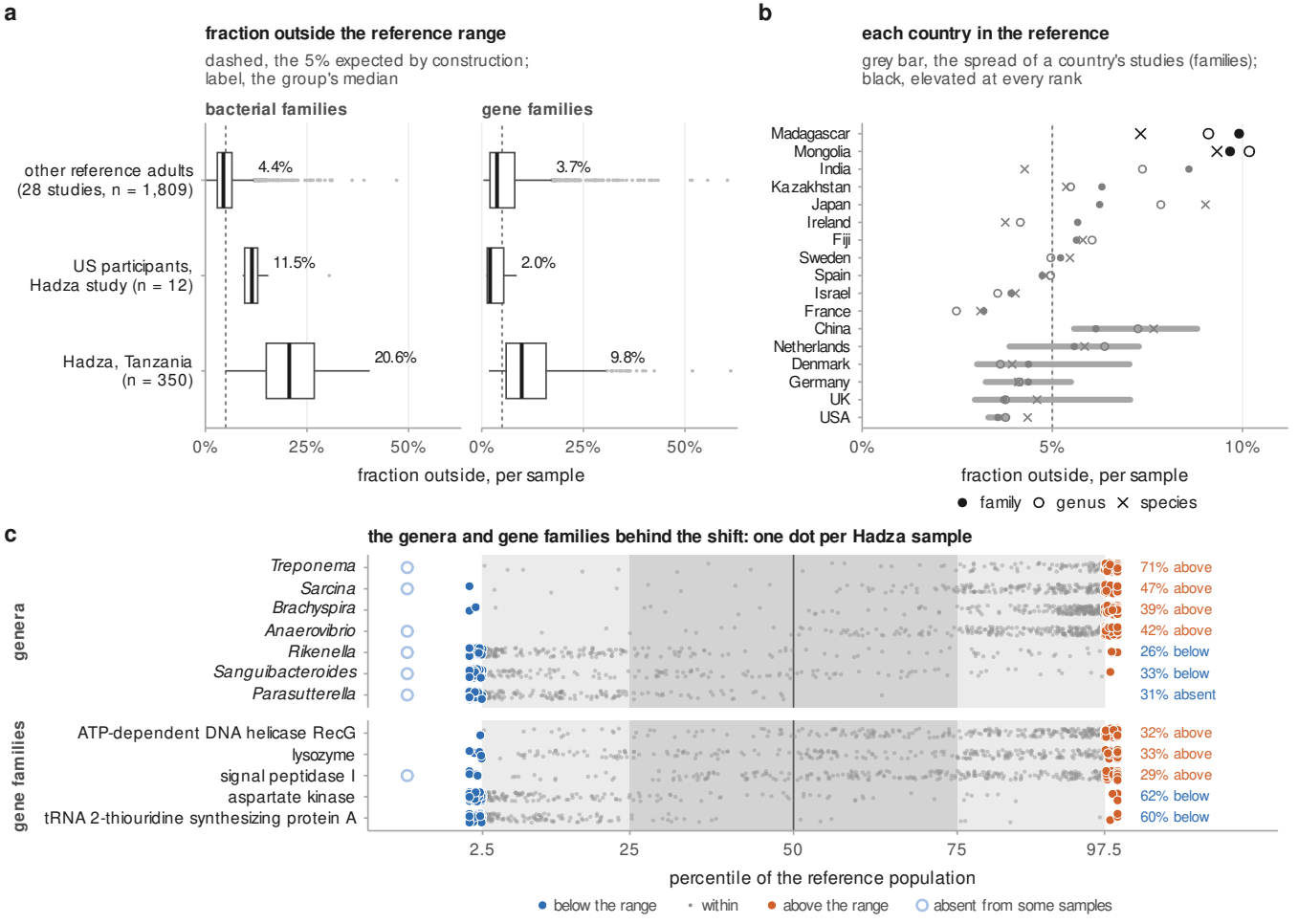

a
fraction outside the reference range
dashed, the 5% expected by construction;
label, the group's median
bacterial families
gene families
other reference adults
(28 studies, n = 1,809)
US participants,
Hadza study (n = 12)
Hadza, Tanzania
(n = 350)
4.4%
11.5%
20.6%
3.7%
2.0%
9.8%
0%
25%
50%
fraction outside, per sample
b
each country in the reference
grey bar, the spread of a country's studies (families);
black, elevated at every rank
Madagascar
Mongolia
India
Kazakhstan
Japan
Ireland
Fiji
Sweden
Spain
Israel
France
China
Netherlands
Denmark
Germany
UK
USA
0%
5%
10%
fraction outside, per sample
family
genus
species
c
the genera and gene families behind the shift: one dot per Hadza sample
genera
Treponema
Sarcina
Brachyspira
Anaerovibrio
Rikenella
Sanguibacteroides
Parasutterella
71% above
47% above
39% above
42% above
26% below
33% below
31% absent
gene families
ATP-dependent DNA helicase RecG
lysozyme
signal peptidase I
aspartate kinase
tRNA 2-thiouridine synthesizing protein A
32% above
33% above
29% above
62% below
60% below
2.5
25
50
75
97.5
percentile of the reference population
below the range
within
above the range
absent from some samples